Projet de Groupe

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Introduction

Dans le cadre du projet, ce rapport présente une analyse approfondie des données de trafic aérien afin d'identifier les causes de dysfonctionnements et d'améliorer la fiabilité du service. S'appuyant sur le croisement des jeux de données flights.xlsx, airlines.json, airports.xlsx et planes.html, cette étude s'articule autour de quatre axes majeurs :

- Exploration et Inventaire : Cartographie du trafic, identification des acteurs clés (aéroports, compagnies, avions) et analyse de la couverture des destinations.
- Dynamique Temporelle : Étude des cycles de trafic (mensuels, hebdomadaires) et identification des pics d'activité pour anticiper la saturation.
- Analyse de la Performance et des Retards : Évaluation de la qualité de service via l'étude des retards au départ et à l'arrivée, calcul des gains de temps en vol (rattrapage) et analyse de la corrélation entre distance, heure de décollage et ponctualité.
- Gestion des Aléas et Qualité des Données : Audit des données manquantes et analyse détaillée des volumes d'annulations par compagnie et par destination.

Ce rapport vise à fournir une vision synthétique et actionnable pour accompagner la prise de décision stratégique.

> Remarque : Ce rapport suppose que les fichiers `flights.xlsx` et `airlines.json` sont disponibles aux chemins suivants sur la machine locale :
>
> - `/Users/nano/Desktop/Exercice/Projet/Data/flights.xlsx`
> - `/Users/nano/Desktop/Exercice/Projet/Data/airlines.json`
> - `/Users/nano/Desktop/Exercice/Projet/Data/airports.xlsx`
> - `/Users/nano/Desktop/Exercice/Projet/Data/planes.html`

0. Lecture et préparation des données

```r
# --- lire les fichiers ---
flights_path <- "/Users/nano/Desktop/Exercice/Projet/Data/flights.xlsx"
airlines_path <- "/Users/nano/Desktop/Exercice/Projet/Data/airlines.json"
airports_path <- "/Users/nano/Desktop/Exercice/Projet/Data/airports.xlsx"
planes_path <- "/Users/nano/Desktop/Exercice/Projet/Data/planes.html"

flights <- read_excel(flights_path)
airports <- read_excel(airports_path)
airlines <- fromJSON(airlines_path)
planes <- read_html(planes_path) %>%
  html_table() %>%
  .[[1]]
```

```r
glimpse(flights)
```

```
## Rows: 252,704
## Columns: 19
## $ year           <dbl> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2~
## $ month          <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ day            <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2~
## $ dep_time       <dbl> 1525, 1528, 1740, 1807, 1939, 1952, 2016, NA, NA, NA, N~
## $ sched_dep_time <dbl> 1530, 1459, 1745, 1738, 1840, 1930, 1930, 1630, 1935, 1~
## $ dep_delay      <dbl> -5, 29, -5, 29, 59, 22, 46, NA, NA, NA, NA, 43, 120, 8,~
## $ arr_time       <dbl> 1934, 2002, 2158, 2251, 29, 2358, NA, NA, NA, NA, NA, 1~
## $ sched_arr_time <dbl> 1805, 1647, 2020, 2103, 2151, 2207, 2220, 1815, 2240, 1~
## $ arr_delay      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ carrier        <chr> "MQ", "EV", "MQ", "UA", "9E", "EV", "EV", "EV", "AA", "~
## $ flight         <dbl> 4525, 3806, 4413, 1228, 3325, 4333, 4204, 4308, 791, 19~
## $ tailnum        <chr> "N719MQ", "N17108", "N739MQ", "N31412", "N905XJ", "N111~
## $ origin         <chr> "LGA", "EWR", "LGA", "EWR", "JFK", "EWR", "EWR", "EWR",~
## $ dest           <chr> "XNA", "STL", "XNA", "SAN", "DFW", "TUL", "OKC", "RDU",~
## $ air_time       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ distance       <dbl> 1147, 872, 1147, 2425, 1391, 1215, 1325, 416, 1389, 109~
## $ hour           <dbl> 15, 14, 17, 17, 18, 19, 19, 16, 19, 15, 6, 8, 9, 18, 17~
## $ minute         <dbl> 30, 59, 45, 38, 40, 30, 30, 30, 35, 0, 0, 22, 25, 40, 2~
## $ time_hour      <chr> "2021-01-01T15:00:00Z", "2021-01-01T14:00:00Z", "2021-0~
```

```r
glimpse(airports)
```

```
## Rows: 1,458
## Columns: 9
## $ ...1  <dbl> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18~
## $ faa   <chr> "04G", "06A", "06C", "06N", "09J", "0A9", "0G6", "0G7", "0P2", "~
## $ name  <chr> "Lansdowne Airport", "Moton Field Municipal Airport", "Schaumbur~
## $ lat   <dbl> 41.13047, 32.46057, 41.98934, 41.43191, 31.07447, 36.37122, 41.4~
## $ lon   <dbl> -80.61958, -85.68003, -88.10124, -74.39156, -81.42778, -82.17342~
## $ alt   <dbl> 1044, 264, 801, 523, 11, 1593, 730, 492, 1000, 108, 409, 875, 10~
## $ tz    <dbl> -5, -6, -6, -5, -5, -5, -5, -5, -5, -8, -5, -6, -5, -5, -5, -5, ~
## $ dst   <chr> "A", "A", "A", "A", "A", "A", "A", "A", "U", "A", "A", "U", "A",~
## $ tzone <chr> "America/New_York", "America/Chicago", "America/Chicago", "Ameri~
```

```r
glimpse(airlines)
```

```
## Rows: 16
## Columns: 2
## $ carrier <chr> "9E", "AA", "AS", "B6", "DL", "EV", "F9", "FL", "HA", "MQ", "O~
## $ name    <chr> "Endeavor Air Inc.", "American Airlines Inc.", "Alaska Airline~
```

```r
glimpse(planes)
```

```
## Rows: 3,322
## Columns: 10
## $ ``           <int> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,~
## $ tailnum      <chr> "N10156", "N102UW", "N103US", "N104UW", "N10575", "N105UW~
## $ year         <int> 2004, 1998, 1999, 1999, 2002, 1999, 1999, 1999, 1999, 199~
## $ type         <chr> "Fixed wing multi engine", "Fixed wing multi engine", "Fi~
## $ manufacturer <chr> "EMBRAER", "AIRBUS INDUSTRIE", "AIRBUS INDUSTRIE", "AIRBU~
## $ model        <chr> "EMB-145XR", "A320-214", "A320-214", "A320-214", "EMB-145~
```

```
## $ engines      <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ~
## $ seats        <int> 55, 182, 182, 182, 55, 182, 182, 182, 182, 182, 55, 55, 5~
## $ speed        <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ engine       <chr> "Turbo-fan", "Turbo-fan", "Turbo-fan", "Turbo-fan", "Turb~
```

1. Statistiques et Exploration Générale

A. Inventaire du Trafic

```r
# Nombre total d'aéroports
total_airports <- unique(c(flights$origin, flights$dest)) %>% length()

# Nombre de compagnies
total_airlines <- n_distinct(flights$carrier)

# Nombre d'avions (tailnum uniques)
total_planes <- n_distinct(flights$tailnum, na.rm = TRUE)

# Nombre de vols potentiellement annulés
total_cancelled_simple <- sum(is.na(flights$dep_time))

cat("Aéroports desservis :", total_airports, "\n")
```

```
## Aéroports desservis : 106
```

```r
cat("Compagnies aériennes :", total_airlines, "\n")
```

```
## Compagnies aériennes : 16
```

```r
cat("Avions distincts :", total_planes, "\n")
```

```
## Avions distincts : 3982
```

```r
cat("Vols sans heure de départ :", total_cancelled_simple, "\n")
```

```
## Vols sans heure de départ : 6481
```

B. Fuseaux Horaires

```r
# Aéroports ne respectant pas l'heure d'été (DST = 'N' pour None, souvent Arizona/Hawaii)
airports_no_dst <- airports %>%
  filter(dst == "N") %>%
  nrow()

# Nombre de fuseaux horaires distincts (colonne tzone)
distinct_tz <- n_distinct(airports$tzone)

cat("Aéroports sans DST ('N') :", airports_no_dst, "\n")
```

```
## Aéroports sans DST ('N') : 23
```

```r
cat("Nombre de fuseaux horaires distincts :", distinct_tz, "\n")
```

```
## Nombre de fuseaux horaires distincts : 10
```

C. Analyse du Trafic : Top et Flop

Aéroport de départ le plus emprunté

```
top_origin <- flights %>%
  count(origin, sort = TRUE) %>%
  slice(1) %>%
  left_join(airports, by = c("origin" = "faa")) %>%
  select(origin, name, n)
top_origin
```

```
## # A tibble: 1 x 3
##   origin name                    n
##   <chr>  <chr>               <int>
## 1 EWR    Newark Liberty Intl 91241
```

Top 10 des Destinations les plus prisées

```
top_dest <- flights %>%
  count(dest, sort = TRUE) %>%
  mutate(percentage = n / sum(n) * 100) %>%
  slice_head(n = 10) %>%
  left_join(airports, by = c("dest" = "faa")) %>%
  select(dest, name, n, percentage)
top_dest
```

```
## # A tibble: 10 x 4
##    dest  name                                   n percentage
##    <chr> <chr>                              <int>      <dbl>
##  1 ATL   Hartsfield Jackson Atlanta Intl    12946       5.12
##  2 ORD   Chicago Ohare Intl                 12654       5.01
##  3 LAX   Los Angeles Intl                   11895       4.71
##  4 BOS   General Edward Lawrence Logan Intl 11560       4.57
##  5 MCO   Orlando Intl                       10637       4.21
##  6 CLT   Charlotte Douglas Intl             10448       4.13
##  7 SFO   San Francisco Intl                  9729       3.85
##  8 FLL   Fort Lauderdale Hollywood Intl      9443       3.74
##  9 MIA   Miami Intl                          8938       3.54
## 10 DCA   Ronald Reagan Washington Natl       7386       2.92
```

Top 10 des Destinations les moins prisées

```
bottom_dest <- flights %>%
  count(dest, sort = TRUE) %>%
  mutate(percentage = n / sum(n) * 100) %>%
  slice_tail(n = 10) %>%
  left_join(airports, by = c("dest" = "faa")) %>%
  select(dest, name, n, percentage)
bottom_dest
```

```
## # A tibble: 10 x 4
##    dest  name                         n percentage
##    <chr> <chr>                    <int>      <dbl>
##  1 CHO   Charlottesville-Albemarle    31    0.0123
##  2 JAC   Jackson Hole Airport         25    0.00989
```

```
##   3 BZN   Gallatin Field               20   0.00791
##   4 PSP   Palm Springs Intl            19   0.00752
##   5 EYW   Key West Intl                17   0.00673
##   6 TVC   Cherry Capital Airport       16   0.00633
##   7 HDN   Yampa Valley                 15   0.00594
##   8 MTJ   Montrose Regional Airport    15   0.00594
##   9 SBN   South Bend Rgnl               4   0.00158
##  10 LEX   Blue Grass                    1   0.000396
```

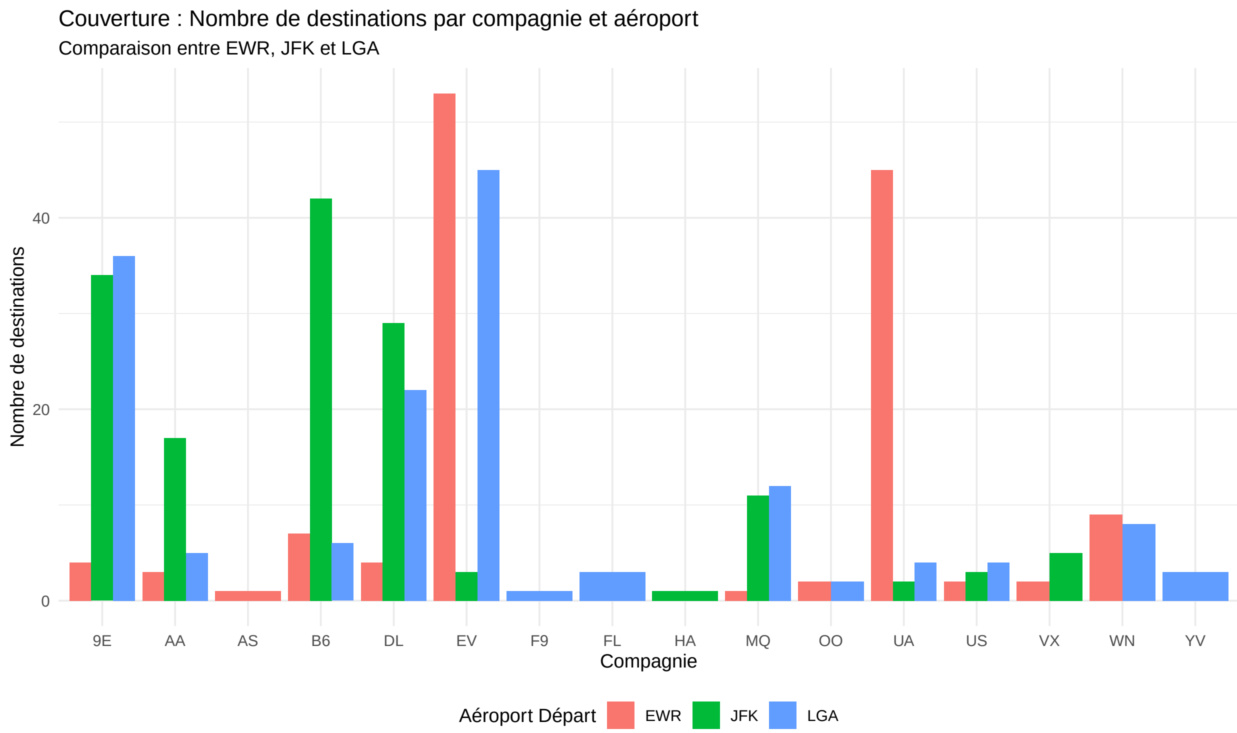
Les 10 Avions les plus actifs

```r
top_planes <- flights %>%
  filter(!is.na(tailnum)) %>%
  count(tailnum, sort = TRUE) %>%
  slice_head(n = 10)
top_planes
```

```
## # A tibble: 10 x 2
##    tailnum     n
##    <chr>   <int>
##  1 N725MQ    443
##  2 N723MQ    394
##  3 N713MQ    385
##  4 N722MQ    378
##  5 N711MQ    376
##  6 N258JB    332
##  7 N353JB    316
##  8 N351JB    310
##  9 N542MQ    310
## 10 N228JB    309
```

D. Couverture des Compagnies

```r
# Préparation des données
dest_per_airline_origin <- flights %>%
  group_by(origin, carrier) %>%
  summarise(n_dest = n_distinct(dest), .groups = 'drop')

# Graphique
ggplot(dest_per_airline_origin, aes(x = carrier, y = n_dest, fill = origin)) +
  geom_col(position = "dodge") +
  labs(title = "Couverture : Nombre de destinations par compagnie et aéroport",
       subtitle = "Comparaison entre EWR, JFK et LGA",
       y = "Nombre de destinations",
       x = "Compagnie",
       fill = "Aéroport Départ") +
  theme_minimal() +
  theme(legend.position = "bottom")
```



Compagnies ne desservant pas l'ensemble des aéroports new-yorkais :

```r
# On compte combien d'aéroports d'origine distincts chaque compagnie utilise.
# Si ce nombre est inférieur au nombre total d'aéroports d'origine (3 : JFK, LGA, EWR)
total_origins <- n_distinct(flights$origin)

airlines_partial_coverage <- flights %>%
  group_by(carrier) %>%
  summarise(n_origins_used = n_distinct(origin)) %>%
  filter(n_origins_used < total_origins)
airlines_partial_coverage
```

```
## # A tibble: 8 x 2
##   carrier n_origins_used
##   <chr>            <int>
## 1 AS                   1
## 2 F9                   1
## 3 FL                   1
## 4 HA                   1
## 5 OO                   2
## 6 VX                   2
## 7 WN                   2
## 8 YV                   1
```

E. Filtres Spécifiques

Vols vers Houston (IAH ou HOU) :

```r
nb_houston <- flights %>%
  filter(dest %in% c("IAH", "HOU")) %>%
  nrow()
cat("Nombre total de vols vers Houston :", nb_houston)
```

```
## Nombre total de vols vers Houston : 6958
```

```r
flights_houston <- flights %>%
  filter(dest %in% c("IAH", "HOU"))
flights_houston
```

```
## # A tibble: 6,958 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <dbl> <dbl> <dbl>    <dbl>          <dbl>     <dbl>    <dbl>          <dbl>
##  1  2021     1     8       NA           1300        NA       NA           1610
##  2  2021     1     9       NA           1251        NA       NA           1602
##  3  2021     1    20      616            625        -9     1203            937
##  4  2021     1    22       NA           1856        NA       NA           2204
##  5  2021     1    30       NA           1300        NA       NA           1610
##  6  2021     1    31       NA           1446        NA       NA           1757
##  7  2021     1    31       NA            625        NA       NA            934
##  8  2021    10     9       NA            630        NA       NA            919
##  9  2021    10    16       NA            720        NA       NA           1000
## 10  2021    10    22       NA            630        NA       NA            919
## # i 6,948 more rows
## # i 11 more variables: arr_delay <lgl>, carrier <chr>, flight <dbl>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <lgl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <chr>
```

Statistiques de la liaison NYC -> Seattle (SEA) :

```r
nyc_sea_stats <- flights %>%
  filter(dest == "SEA") %>%
  summarise(
    nb_vols = n(),
    nb_compagnies = n_distinct(carrier),
    nb_avions = n_distinct(tailnum)
  )
nyc_sea_stats
```

```
## # A tibble: 1 x 3
##   nb_vols nb_compagnies nb_avions
##     <int>         <int>     <int>
## 1    2736             5       857
```

```r
# Destinations exclusives à certaines compagnies
exclusive_dest <- flights %>%
  group_by(dest) %>%
  distinct(dest, carrier) %>%
  left_join(airports, by = c("dest" = "faa")) %>%
  left_join(airlines, by = "carrier") %>%
  select(Destination = dest, Compagnie = name.y) %>%
  arrange(Destination)
exclusive_dest
```

```
## # A tibble: 308 x 2
## # Groups:   Destination [103]
##    Destination Compagnie
##    <chr>       <chr>
##  1 ABQ         JetBlue Airways
##  2 ACK         JetBlue Airways
##  3 ALB         ExpressJet Airlines Inc.
##  4 ATL         ExpressJet Airlines Inc.
##  5 ATL         Endeavor Air Inc.
##  6 ATL         Delta Air Lines Inc.
##  7 ATL         AirTran Airways Corporation
##  8 ATL         Envoy Air
##  9 ATL         Southwest Airlines Co.
## 10 ATL         United Air Lines Inc.
## # i 298 more rows
```

Vols majeurs (UA, AA, DL) :

```r
nb_major <- flights %>%
  filter(carrier %in% c("UA", "AA", "DL")) %>%
  nrow()
part_marche <- round(nb_major / nrow(flights) * 100, 1)
cat("Les compagnies United, American et Delta représentent",
    nb_major, "vols, soit", part_marche, "% du trafic total.")
```

```
## Les compagnies United, American et Delta représentent 104742 vols, soit 41.4 % du trafic total.
```

2. Analyse des Retards et Fiabilité

A. Reconstitution des Datetimes Complets

```r
# Convertit HHMM en datetime
to_datetime <- function(year, month, day, time_col) {
  time_str <- ifelse(is.na(time_col), NA, sprintf("%04d", time_col))
  hour <- ifelse(is.na(time_str), NA_integer_, as.integer(substr(time_str, 1, 2)))
  minute <- ifelse(is.na(time_str), NA_integer_, as.integer(substr(time_str, 3, 4)))
  lubridate::make_datetime(
    year  = year,
    month = month,
    day   = day,
    hour  = hour,
    min   = minute,
    sec   = 0
  )
}

flights2 <- flights %>%
  mutate(
    sched_dep_datetime = to_datetime(year, month, day, sched_dep_time),
    dep_datetime       = to_datetime(year, month, day, dep_time),
    arr_datetime       = to_datetime(year, month, day, arr_time),
    sched_arr_datetime = to_datetime(year, month, day, sched_arr_time),
    date = as_date(sched_dep_datetime)
```

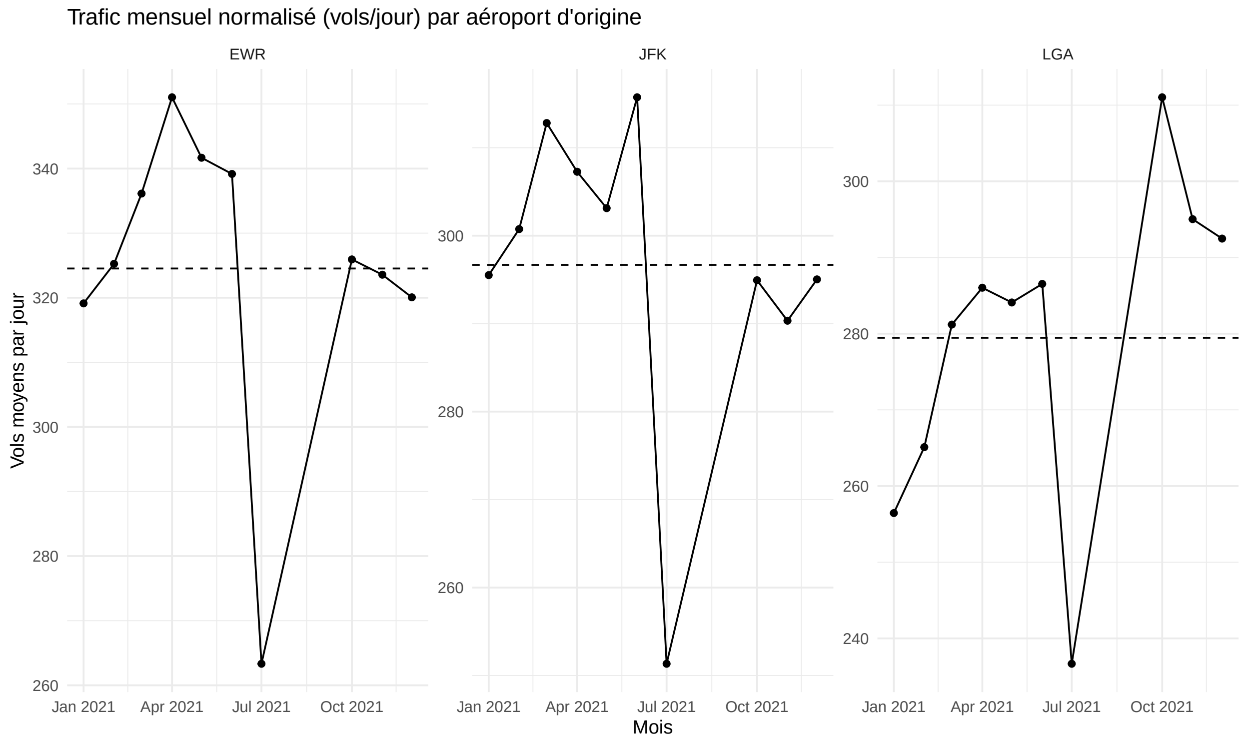
```r
  ) %>%
  select(
    -year, -month, -day, -hour, -minute
  )
flights2["date"]
```

```
## # A tibble: 252,704 x 1
##    date
##    <date>
##  1 2021-01-01
##  2 2021-01-01
##  3 2021-01-01
##  4 2021-01-01
##  5 2021-01-01
##  6 2021-01-01
##  7 2021-01-01
##  8 2021-01-01
##  9 2021-01-01
## 10 2021-01-01
## # i 252,694 more rows
```

B. Trafic Mensuel Normalisé (Vols/Jour)

```r
daily_traffic <- flights2 %>%
  group_by(origin, date) %>%
  summarise(flights_per_day = n(), .groups = "drop")

monthly_traffic <- daily_traffic %>%
  mutate(month = floor_date(date, "month")) %>%
  group_by(origin, month) %>%
  summarise(
    flights_total = sum(flights_per_day),
    days_in_month = n_distinct(date),
    flights_per_day_mean = flights_total / days_in_month,
    .groups = "drop"
  )

origin_monthly_mean <- monthly_traffic %>%
  group_by(origin) %>%
  summarise(origin_mean_flights_per_day = mean(flights_per_day_mean))

monthly_traffic <- monthly_traffic %>%
  left_join(origin_monthly_mean, by = "origin")

# Visualisation facettée
p_monthly <- ggplot(monthly_traffic, aes(x = month, y = flights_per_day_mean)) +
  geom_line() +
  geom_point() +
  geom_hline(
    data = origin_monthly_mean,
    aes(yintercept = origin_mean_flights_per_day),
    linetype = "dashed"
  ) +
```

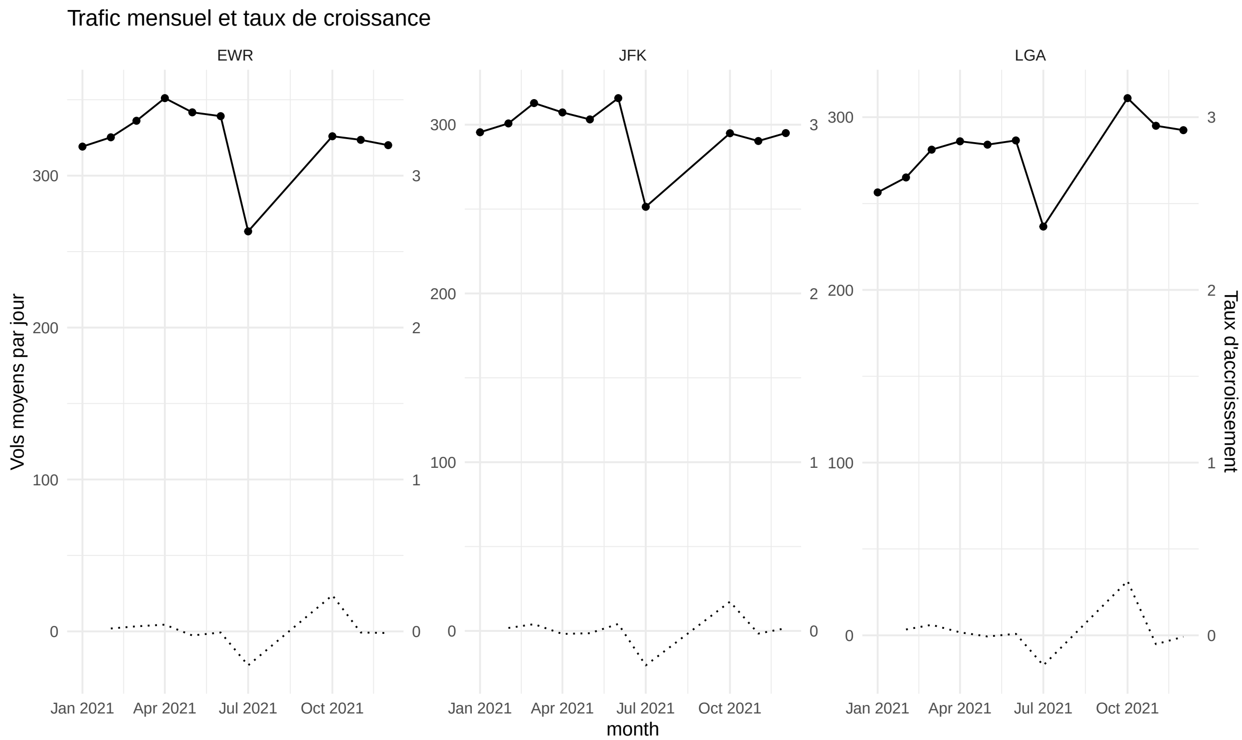
```r
  facet_wrap(~ origin, scales = "free_y") +
  labs(
    title = "Trafic mensuel normalisé (vols/jour) par aéroport d'origine",
    x = "Mois",
    y = "Vols moyens par jour"
  ) +
  theme_minimal()

print(p_monthly)
```



C. Taux d'Accroissement Mensuel

```r
monthly_growth <- monthly_traffic %>%
  arrange(origin, month) %>%
  group_by(origin) %>%
  mutate(
    growth_rate = (flights_per_day_mean - lag(flights_per_day_mean)) /
      lag(flights_per_day_mean)
  ) %>%
  ungroup()

scale_factor <- 100

p_monthly_growth <- ggplot(monthly_growth, aes(x = month)) +
  geom_line(aes(y = flights_per_day_mean)) +
  geom_point(aes(y = flights_per_day_mean)) +
  geom_line(aes(y = growth_rate * scale_factor), linetype = "dotted") +
  facet_wrap(~ origin, scales = "free_y") +
  scale_y_continuous(
```

```r
    name = "Vols moyens par jour",
    sec.axis = sec_axis(~ . / scale_factor, name = "Taux d'accroissement")
  ) +
  labs(title = "Trafic mensuel et taux de croissance") +
  theme_minimal()

print(p_monthly_growth)
```

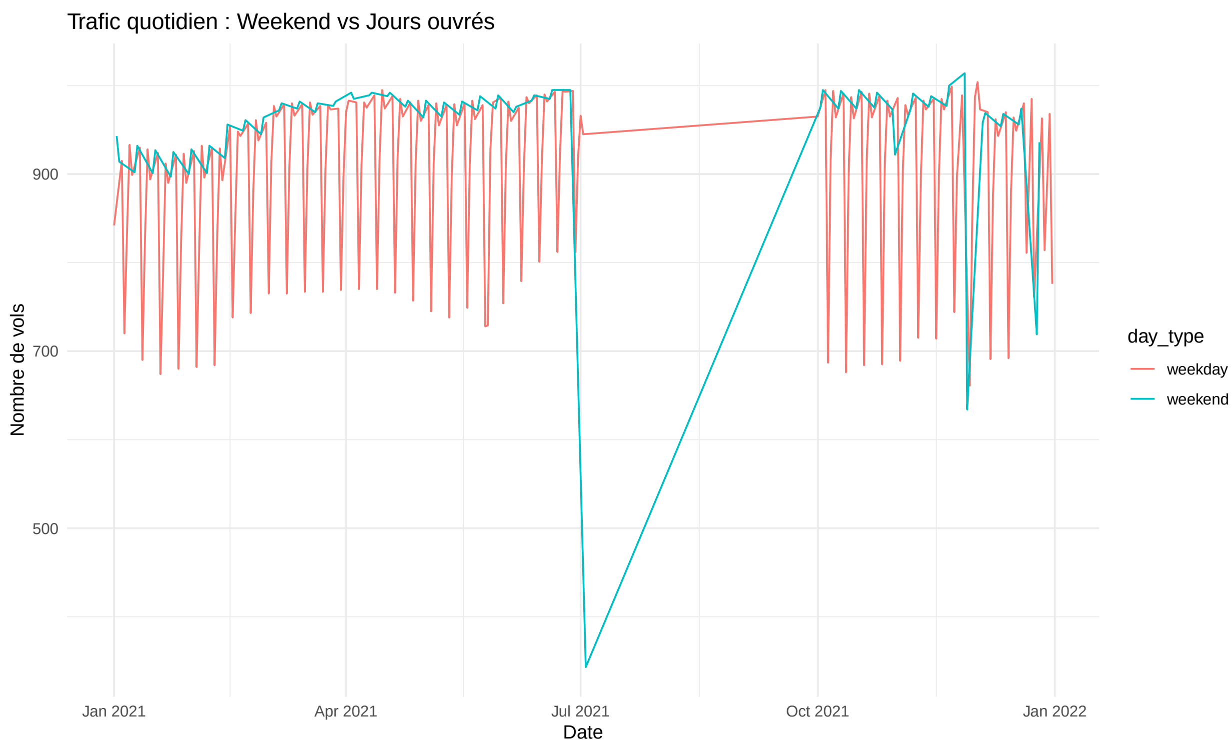


D. Weekend vs Jours Ouvrés

```r
weekly_traffic <- flights2 %>%
  mutate(
    wday = wday(date, week_start = 1),
    day_type = if_else(wday %in% c(6, 7), "weekend", "weekday")
  )

daily_by_daytype <- weekly_traffic %>%
  group_by(day_type, date) %>%
  summarise(flights = n(), .groups = "drop")

daytype_stats <- daily_by_daytype %>%
  group_by(day_type) %>%
  summarise(
    mean_flights = mean(flights),
    sd_flights = sd(flights)
  )

print(daytype_stats)
```

```
## # A tibble: 2 x 3
##   day_type mean_flights sd_flights
##   <chr>           <dbl>      <dbl>
## 1 weekday          901.       96.5
## 2 weekend          951.       87.7
```

```r
p_weekday <- ggplot(daily_by_daytype, aes(x = date, y = flights, colour = day_type)) +
  geom_line() +
  theme_minimal() +
  labs(
    title = "Trafic quotidien : Weekend vs Jours ouvrés",
    x = "Date",
    y = "Nombre de vols"
  )

print(p_weekday)
```



E. Jours Spéciaux vs Moyenne Annuelle

```r
holidays_2021 <- as.Date(c(
  "2021-01-01",
  "2021-07-04",
  "2021-11-25",
  "2021-12-25"
))

flights_special <- flights2 %>%
  mutate(is_special_day = date %in% holidays_2021)

special_daily <- flights_special %>%
```

```r
  group_by(is_special_day, date) %>%
  summarise(flights = n(), .groups = "drop")

special_stats <- special_daily %>%
  group_by(is_special_day) %>%
  summarise(mean_flights = mean(flights))

print(special_stats)
```

```
## # A tibble: 2 x 2
##   is_special_day mean_flights
##   <lgl>                 <dbl>
## 1 FALSE                  916.
## 2 TRUE                   834.
```

```r
annual_mean <- flights2 %>%
  group_by(date) %>%
  summarise(flights = n()) %>%
  summarise(annual_mean = mean(flights))

print(annual_mean)
```

```
## # A tibble: 1 x 1
##   annual_mean
##         <dbl>
## 1        916.
```

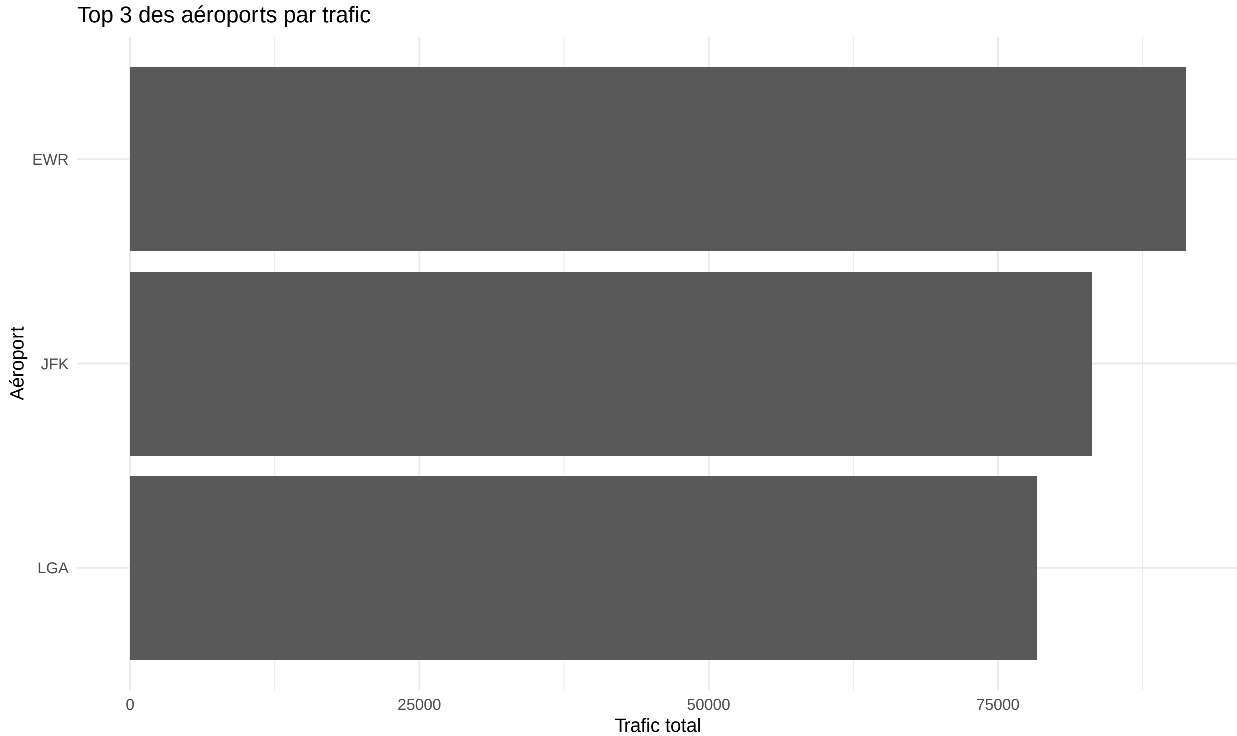
F. Top 3 Aéroports

```r
origin_counts <- flights2 %>% count(airport = origin, name = "n_origin")
dest_counts   <- flights2 %>% count(airport = dest,   name = "n_dest")

airport_traffic <- full_join(origin_counts, dest_counts, by = "airport") %>%
  replace_na(list(n_origin = 0, n_dest = 0)) %>%
  mutate(total_traffic = n_origin + n_dest)

top3 <- airport_traffic %>%
  arrange(desc(total_traffic)) %>%
  slice_head(n = 3)

print(top3)
```

```
## # A tibble: 3 x 4
##   airport n_origin n_dest total_traffic
##   <chr>      <int>  <int>         <int>
## 1 EWR        91241      0         91241
## 2 JFK        83119      0         83119
## 3 LGA        78344      0         78344
```

```r
p_top3 <- ggplot(top3, aes(x = reorder(airport, total_traffic), y = total_traffic)) +
  geom_col() +
  coord_flip() +
  labs(title = "Top 3 des aéroports par trafic", x = "Aéroport", y = "Trafic total") +
  theme_minimal()
```

```r
print(p_top3)
```



G. Fonctions Utilitaires et Calculs de Retard (Gestion du Passage Minuit)

```r
# Convertir hhmm (ex: 524, 1330) en minutes depuis minuit
hhmm_to_minutes <- function(time) {
  time <- as.numeric(time)
  hour <- floor(time / 100)
  minute <- time %% 100
  out <- hour * 60 + minute
  out[is.na(time)] <- NA
  return(out)
}

# Calcul des retards arr_delay et dep_delay (si dep_delay déjà présent, on l'utilise)
# Pour arr_delay on recalcule à partir d'arr_time et
# sched_arr_time pour corriger passage minuit

df2 <- flights %>%
  mutate(
    arr_time_min   = ifelse(is.na(arr_time), NA_real_,
                        hhmm_to_minutes(arr_time)),
    sched_arr_min  = ifelse(is.na(sched_arr_time), NA_real_,
                        hhmm_to_minutes(sched_arr_time)),
    dep_time_min   = ifelse(is.na(dep_time), NA_real_,
                        hhmm_to_minutes(dep_time)),
    sched_dep_min  = ifelse(is.na(sched_dep_time), NA_real_,
                        hhmm_to_minutes(sched_dep_time))
```

```r
  ) %>%
  rowwise() %>%
  mutate(
    arr_delay_calc = if(any(is.na(c(arr_time_min, sched_arr_min)))) NA_real_ else {
      delta <- arr_time_min - sched_arr_min
      if(delta < -720) delta <- delta + 1440
      if(delta > 720) delta <- delta - 1440
      delta
    },
    dep_delay_calc = if(!is.na(dep_delay)) dep_delay
    else if(any(is.na(c(dep_time_min, sched_dep_min)))) NA_real_ else {
      delta2 <- dep_time_min - sched_dep_min
      if(delta2 < -720) delta2 <- delta2 + 1440
      if(delta2 > 720) delta2 <- delta2 - 1440
      delta2
    }
  ) %>%
  ungroup()

# Remplacer les colonnes arr_delay/dep_delay originales par les calculées
# si plus cohérentes
df2 <- df2 %>%
  rename(arr_delay_raw = arr_delay, dep_delay_raw = dep_delay) %>%
  mutate(
    arr_delay = arr_delay_calc,
    dep_delay = dep_delay_calc
  ) %>%
  select(-arr_time_min, -sched_arr_min, -dep_time_min, -sched_dep_min,
         -arr_delay_calc, -dep_delay_calc)

# Joindre les informations compagnies
if(!("carrier" %in% names(airlines))) {
  names(airlines) <- c("carrier", "name")[1:ncol(airlines)]
}

df2 <- df2 %>% left_join(airlines, by = "carrier")

cat("Données préparées - lignes:", nrow(df2), "\n")
```

```
## Données préparées - lignes: 252704
```

H. Table "master" avec Indicateurs de Retard, Gain et Rattrapage

```r
df_master <- df2 %>%
  mutate(
    delayed_arr = ifelse(is.na(arr_delay), NA, arr_delay > 0),
    delayed_dep = ifelse(is.na(dep_delay), NA, dep_delay > 0),
    early_arr   = ifelse(is.na(arr_delay), NA, arr_delay < 0),
    early_dep   = ifelse(is.na(dep_delay), NA, dep_delay < 0),
    gain_vol    = ifelse(is.na(dep_delay) | is.na(arr_delay), NA, dep_delay - arr_delay),
    # gain_per_hour = gain en minutes divisé par air_time en heures
    gain_per_hour = ifelse(is.na(gain_vol) | is.na(air_time) |
                             air_time <= 0, NA, gain_vol / (air_time/60)),
```

```r
    rattrape    = ifelse(is.na(arr_delay) | is.na(dep_delay), NA, arr_delay < dep_delay)
  )

# Petite vérification pour voir si les valeurs sont cohérentes et
# supprimer les valeurs extremes
summary(select(df_master, arr_delay, dep_delay, gain_vol, gain_per_hour, air_time))
```

```
##    arr_delay          dep_delay          gain_vol        gain_per_hour
##  Min.    :-720.000   Min.   : -43.00   Min.   :-547.000   Min.   :-11760.0
##  1st Qu.: -16.000   1st Qu.:  -5.00   1st Qu.:  -4.000   1st Qu.:  -240.0
##  Median :  -4.000   Median :  -2.00   Median :   7.000   Median :   420.0
##  Mean   :   7.438   Mean   :  12.41   Mean   :   4.922   Mean   :   313.8
##  3rd Qu.:  15.000   3rd Qu.:  11.00   3rd Qu.:  16.000   3rd Qu.:   960.0
##  Max.   : 688.000   Max.   :1301.00   Max.   :1489.000   Max.   : 89340.0
##  NA's   :6767       NA's   :6481      NA's   :6767       NA's   :7273
##  air_time
##  Mode:logical
##  TRUE:245431
##  NA's:7273
##
##
##
##
```

I. Filtrage des Valeurs Aberrantes

```r
# Filtrage raisonnable pour les analyses graphiques
df_filtered <- df_master %>%
  filter(!is.na(arr_delay), !is.na(dep_delay),
         arr_delay >= -30, arr_delay <= 1440,  # tolérer quelques cas extrêmes
         dep_delay >= -30, dep_delay <= 1440,
         !is.na(distance), distance > 0,
         !is.na(air_time), air_time > 0)

cat("Après filtrage - lignes:", nrow(df_filtered), "\n")
```

```
## Après filtrage - lignes: 232083
```

J. Fiabilité/Régularité par Compagnie (Tableau et Ranking)

```r
fiabilite <- df_filtered %>%
  group_by(carrier, name) %>%
  summarise(
    n = n(),
    prop_delayed_arr = mean(delayed_arr, na.rm = TRUE),
    prop_delayed_dep = mean(delayed_dep, na.rm = TRUE),
    prop_early_arr   = mean(early_arr, na.rm = TRUE),
    prop_early_dep   = mean(early_dep, na.rm = TRUE),
    mean_arr_delay   = mean(arr_delay, na.rm = TRUE),
    sd_arr_delay     = sd(arr_delay, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(mean_arr_delay)
```

```r
# Affichage
fiabilite %>% head(20) %>% kable(digits = 2) %>% kable_styling(full_width = FALSE)
```

carrier	name	n	prop_delayed_arr	prop_delayed_dep	prop_early_arr	prop_early
HA	Hawaiian Airlines Inc.	192	0.32	0.26	0.67	
US	US Airways Inc.	14532	0.39	0.24	0.59	
AS	Alaska Airlines Inc.	420	0.39	0.40	0.59	
DL	Delta Air Lines Inc.	32697	0.37	0.33	0.61	
AA	American Airlines Inc.	21620	0.39	0.35	0.59	
VX	Virgin America	3204	0.40	0.49	0.57	
UA	United Air Lines Inc.	40413	0.43	0.50	0.55	
B6	JetBlue Airways	39041	0.46	0.40	0.52	
MQ	Envoy Air	18520	0.48	0.32	0.50	
WN	Southwest Airlines Co.	8650	0.47	0.54	0.51	
9E	Endeavor Air Inc.	12073	0.43	0.43	0.56	
YV	Mesa Airlines Inc.	363	0.48	0.46	0.50	
FL	AirTran Airways Corporation	2417	0.60	0.51	0.38	
EV	ExpressJet Airlines Inc.	37425	0.52	0.48	0.46	
F9	Frontier Airlines Inc.	508	0.59	0.43	0.38	
OO	SkyWest Airlines Inc.	8	0.50	0.62	0.50	

```r
# Classement par retard moyen croissant
fiabilite_ranked <- fiabilite %>% arrange(mean_arr_delay)
```

K. Efficacité du Vol (le Temps de Rattrapage de Retard)

```r
efficacite <- df_filtered %>%
  group_by(carrier, name) %>%
  summarise(
    n = n(),
    mean_gain_vol   = mean(gain_vol, na.rm = TRUE),
    median_gain_vol = median(gain_vol, na.rm = TRUE),
    sd_gain_vol     = sd(gain_vol, na.rm = TRUE),
    mean_gain_per_h = mean(gain_per_hour, na.rm = TRUE),
    prop_vol_rattrape = mean(rattrape, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(mean_gain_vol))

# Table
kable(efficacite %>% head(20), digits = 2) %>% kable_styling(full_width = FALSE)
```

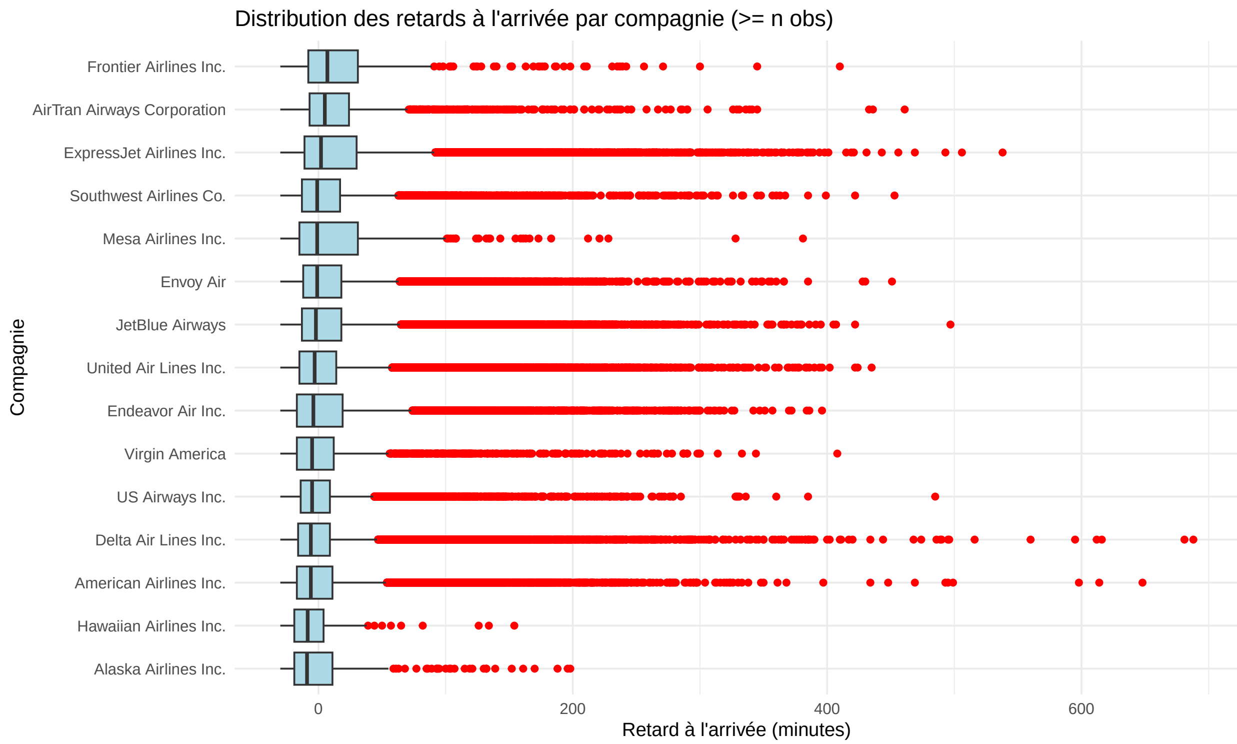
carrier	name	n	mean_gain_vol	median_gain_vol	sd_gain_vol	mean_gain_per_
AS	Alaska Airlines Inc.	420	7.84	9.5	17.70	470.4
VX	Virgin America	3204	6.76	9.0	18.75	405.7
9E	Endeavor Air Inc.	12073	6.18	9.0	17.12	370.6
WN	Southwest Airlines Co.	8650	6.14	8.0	15.73	368.3
HA	Hawaiian Airlines Inc.	192	6.12	7.0	19.74	367.5

UA	United Air Lines Inc.	40413	5.65	8.0	17.56	339.1
DL	Delta Air Lines Inc.	32697	4.94	7.0	16.84	296.1
AA	American Airlines Inc.	21620	4.60	7.0	17.55	276.0
YV	Mesa Airlines Inc.	363	3.60	6.0	15.18	215.8
EV	ExpressJet Airlines Inc.	37425	2.56	5.0	14.40	153.3
B6	JetBlue Airways	39041	2.04	5.0	16.15	122.4
US	US Airways Inc.	14532	0.50	3.0	15.20	29.7
MQ	Envoy Air	18520	-1.03	2.0	16.21	-61.8
FL	AirTran Airways Corporation	2417	-2.01	0.0	15.23	-120.6
F9	Frontier Airlines Inc.	508	-2.84	0.5	20.00	-170.6
OO	SkyWest Airlines Inc.	8	-5.75	-1.5	19.60	-345.0

L. Visualisations — Classement & Distribution

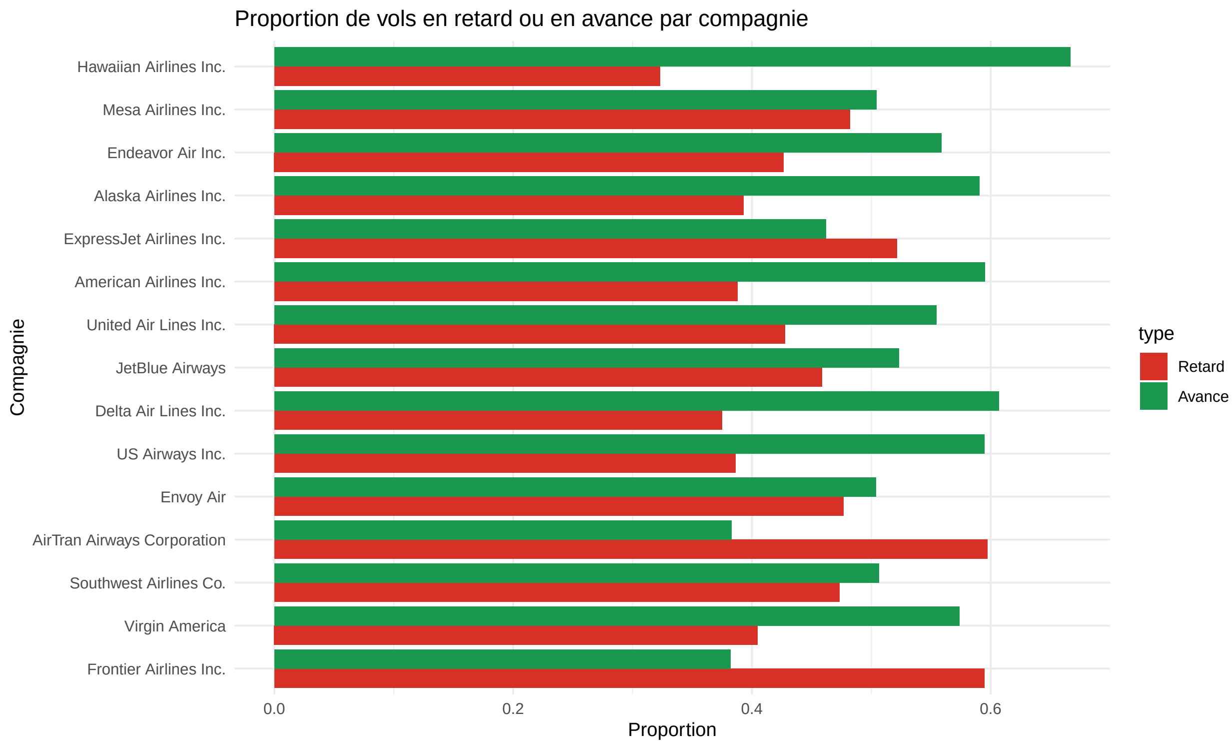
Boxplot des Retards à l'Arrivée par Compagnie (Valeurs Raisonnables)

```r
# On restreint aux compagnies avec au moins un seuil minimal d'observations
# pour lisibilité
min_obs <- 100
top_companies <- fiabilite %>% filter(n >= min_obs) %>% arrange(mean_arr_delay) %>%
  pull(name)

p1 <- ggplot(df_filtered %>% filter(name %in% top_companies),
             aes(x = reorder(name, arr_delay, FUN = median), y = arr_delay)) +
  geom_boxplot(outlier.colour = "red", fill = "lightblue") +
  labs(title = "Distribution des retards à l'arrivée par compagnie (>= n obs)",
       x = "Compagnie", y = "Retard à l'arrivée (minutes)") +
  theme_minimal() +
  coord_flip()

print(p1)
```



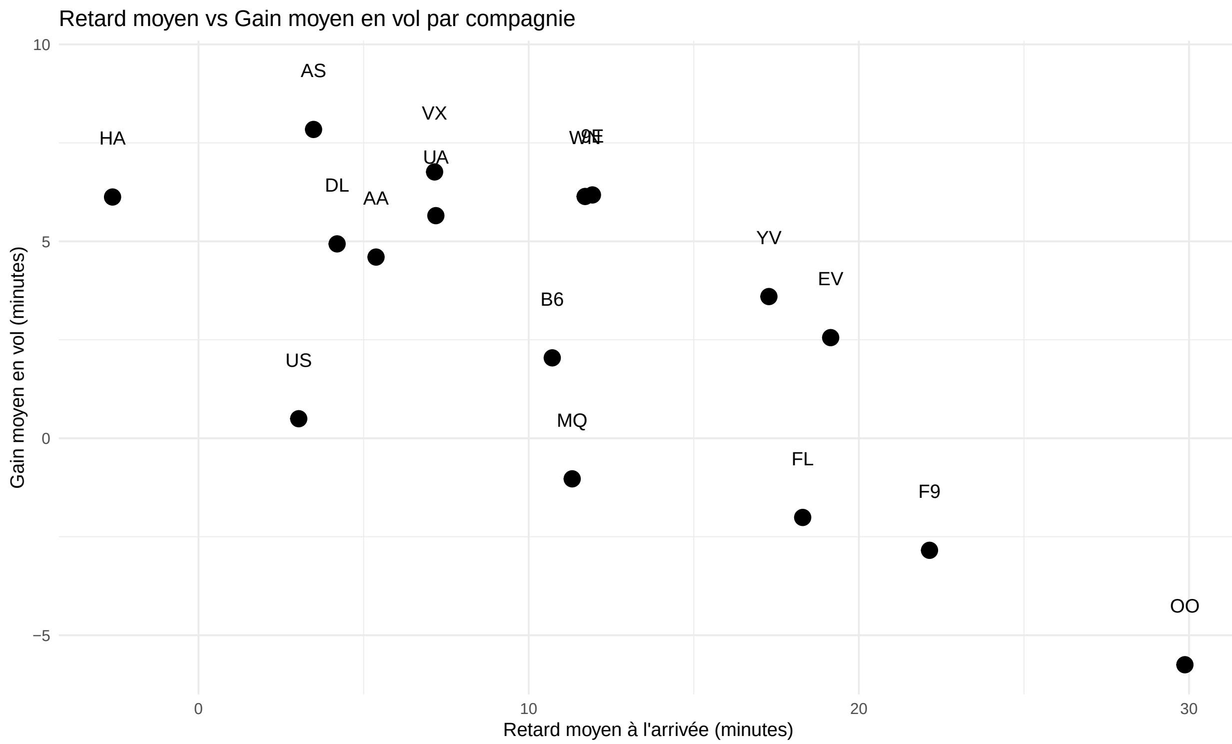
Barplot : Proportion Vols Retardés vs en Avance (par Compagnie)

```
fiabilite_plot <- fiabilite %>% filter(n >= min_obs) %>%
  select(name, prop_delayed_arr, prop_early_arr) %>%
  pivot_longer(cols = c(prop_delayed_arr, prop_early_arr), names_to = "type",
               values_to = "proportion")

p2 <- ggplot(fiabilite_plot, aes(x = reorder(name, proportion),
                                 y = proportion, fill = type)) +
  geom_col(position = "dodge") +
  labs(title = "Proportion de vols en retard ou en avance par compagnie",
       x = "Compagnie", y = "Proportion") +
  scale_fill_manual(values = c("prop_delayed_arr" = "#d73027",
                               "prop_early_arr" = "#1a9850"),
                    labels = c("Retard", "Avance")) +
  theme_minimal() +
  coord_flip()

print(p2)
```



Scatter : Retard Moyen vs Gain Moyen (par Compagnie)

```r
plot_df <- fiabilite %>% select(carrier, name, mean_arr_delay) %>%
  left_join(efficacite %>% select(carrier, mean_gain_vol), by = "carrier") %>%
  distinct()

p3 <- ggplot(plot_df, aes(x = mean_arr_delay, y = mean_gain_vol, label = carrier)) +
  geom_point(size = 4) +
  geom_text(nudge_y = 1.5) +
  labs(title = "Retard moyen vs Gain moyen en vol par compagnie",
       x = "Retard moyen à l'arrivée (minutes)", y = "Gain moyen en vol (minutes)") +
  theme_minimal()

print(p3)
```



M. Calculs de Gain et Identification des Vols ayant Rattrapé le Retard

```r
# Vols qui ont rattrapé : arr_delay < dep_delay (gain_vol positif)
vols_rattrape <- df_filtered %>% filter(!is.na(gain_vol) & gain_vol > 0 & gain_vol < 1000)
cat("Nombre de vols qui ont rattrapé leur retard:", nrow(vols_rattrape), "\n")
```

```
## Nombre de vols qui ont rattrapé leur retard: 149189
```

```r
# Top 10 vols qui ont le plus rattrapé
vols_rattrape %>% arrange(desc(gain_vol)) %>%
  select(year, month, day, carrier, flight, origin, dest, dep_delay,
         arr_delay, gain_vol, gain_per_hour, air_time) %>%
  head(10) %>% kable(digits = 1) %>% kable_styling()
```

year	month	day	carrier	flight	origin	dest	dep_delay	arr_delay	gain_vol	gain_per_hour	air_time
2021	6	13	EV	4377	EWR	JAX	235	126	109	6540	TRUE
2021	2	26	HA	51	JFK	HNL	60	-27	87	5220	TRUE
2021	2	23	HA	51	JFK	HNL	206	126	80	4800	TRUE
2021	3	1	AA	1589	EWR	DFW	368	298	70	4200	TRUE
2021	5	6	VX	183	EWR	SFO	186	117	69	4140	TRUE
2021	3	1	AA	181	JFK	LAX	164	96	68	4080	TRUE
2021	3	1	AA	21	JFK	LAX	184	118	66	3960	TRUE
2021	7	2	UA	356	EWR	PDX	105	39	66	3960	TRUE
2021	3	1	AA	133	JFK	LAX	113	48	65	3900	TRUE
2021	3	1	US	35	JFK	PHX	127	63	64	3840	TRUE

N. Relation Distance - Retard par destination

```r
# Calcul par destination
dest_stats <- df_filtered %>%
  group_by(dest) %>%
  summarise(
    n = n(),
    mean_arr_delay = mean(arr_delay, na.rm = TRUE),
    sd_arr_delay   = sd(arr_delay, na.rm = TRUE),
    mean_distance  = mean(distance, na.rm = TRUE),
    .groups = "drop"
  )

# Identifier HNL (si présent) et vérifier valeurs extrêmes
dest_stats %>% arrange(desc(mean_distance)) %>% head(10) %>% kable(digits = 1) %>%
  kable_styling()
```

dest	n	mean_arr_delay	sd_arr_delay	mean_distance
HNL	437	5.1	38.7	4971.8
SFO	8208	7.8	41.2	2578.2
OAK	189	7.3	37.7	2576.0
SJC	211	4.8	36.7	2569.0
SMF	179	13.9	40.3	2521.0
LAX	10435	5.7	36.9	2468.8
BUR	256	9.8	38.2	2465.0
LGB	429	5.8	36.0	2465.0
PDX	797	8.4	39.5	2445.8
SAN	1749	8.3	35.6	2437.3

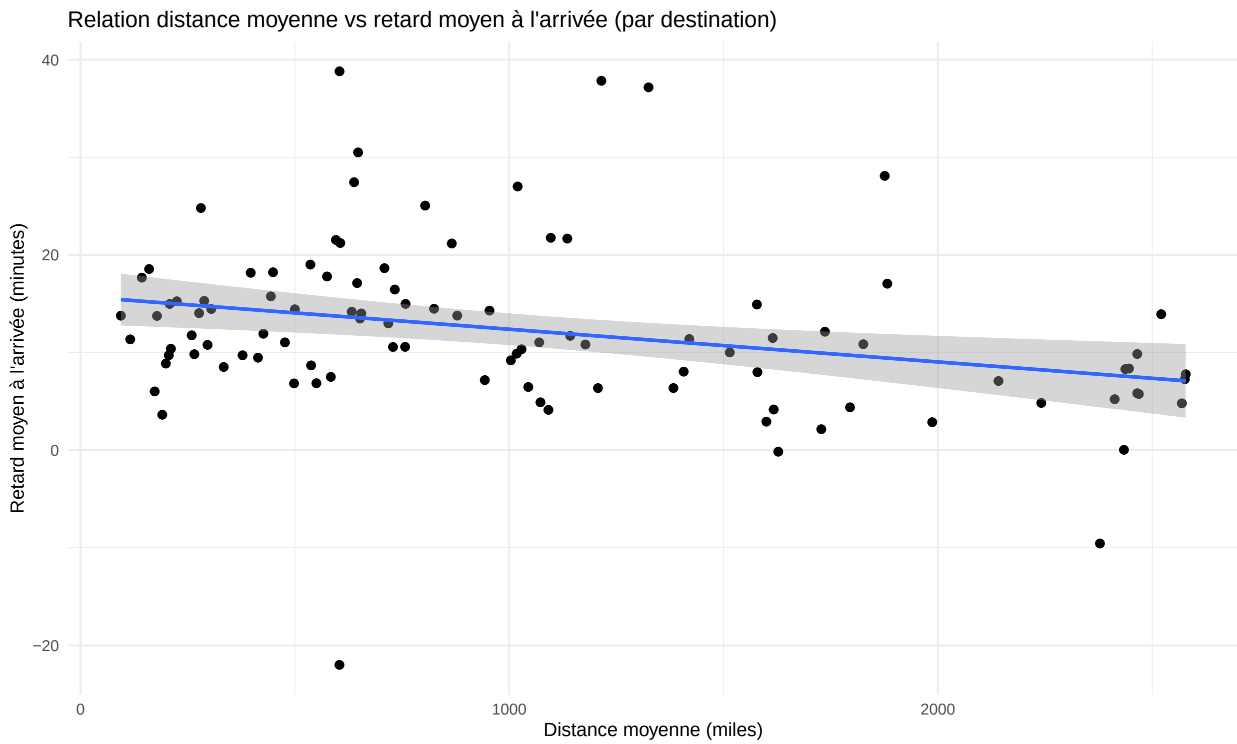
```r
# Exclure HNL si outlier présent
if("HNL" %in% dest_stats$dest) {
  dest_stats_noHNL <- dest_stats %>% filter(dest != "HNL")
} else {
  dest_stats_noHNL <- dest_stats
}

# Corrélation et regression
cor_test <- cor.test(dest_stats_noHNL$mean_distance, dest_stats_noHNL$mean_arr_delay)
lm1 <- lm(mean_arr_delay ~ mean_distance, data = dest_stats_noHNL)
summary_lm1 <- summary(lm1)

# Résumé
cat("Nombre de destinations analysées:", nrow(dest_stats_noHNL), "\n")
```

```
## Nombre de destinations analysées: 102
```

```r
cat("Corrélation (Pearson) entre distance moyenne et retard moyen:",
    round(cor_test$estimate, 3),
    " (p=", signif(cor_test$p.value, 3), ")\n")
```

```
## Corrélation (Pearson) entre distance moyenne et retard moyen: -0.288  (p= 0.00337 )
```

```r
# Graphique scatter + trendline
p4 <- ggplot(dest_stats_noHNL, aes(x = mean_distance, y = mean_arr_delay)) +
  geom_point(size = 2) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(title = "Relation distance moyenne vs retard moyen à l'arrivée (par destination)",
       x = "Distance moyenne (miles)", y = "Retard moyen à l'arrivée (minutes)") +
  theme_minimal()

print(p4)
```




```r
# Afficher résumé du modèle
summary_lm1
```

```
##
## Call:
## lm(formula = mean_arr_delay ~ mean_distance, data = dest_stats_noHNL)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -35.711  -4.751  -0.712   2.405  26.172
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   15.734518   1.420508  11.077  < 2e-16 ***
## mean_distance -0.003350   0.001115  -3.003  0.00337 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.278 on 100 degrees of freedom
## Multiple R-squared:  0.08274,    Adjusted R-squared:  0.07356
```

```
## F-statistic:  9.02 on 1 and 100 DF,  p-value: 0.003374
```

O. Quel est l'Aéroport avec le Retard Moyen le plus Faible ?

```r
# On récupère le meilleur, puis on joint avec la table 'airports' pour avoir le nom
best_dest <- dest_stats %>%
  arrange(mean_arr_delay) %>%
  slice(1) %>%
  left_join(airports, by = c("dest" = "faa")) # <-- C'est ici qu'on récupère le nom

# Affichage du tableau avec le nom
best_dest %>%
  select(Code = dest, Aéroport = name, Vols = n, Retard_Moyen = mean_arr_delay) %>%
  knitr::kable(digits = 2, caption = "L'aéroport de destination le plus fiable") %>%
  kableExtra::kable_styling()
```

Table 5: L'aéroport de destination le plus fiable

Code	Aéroport	Vols	Retard_Moyen
LEX	Blue Grass	1	-22

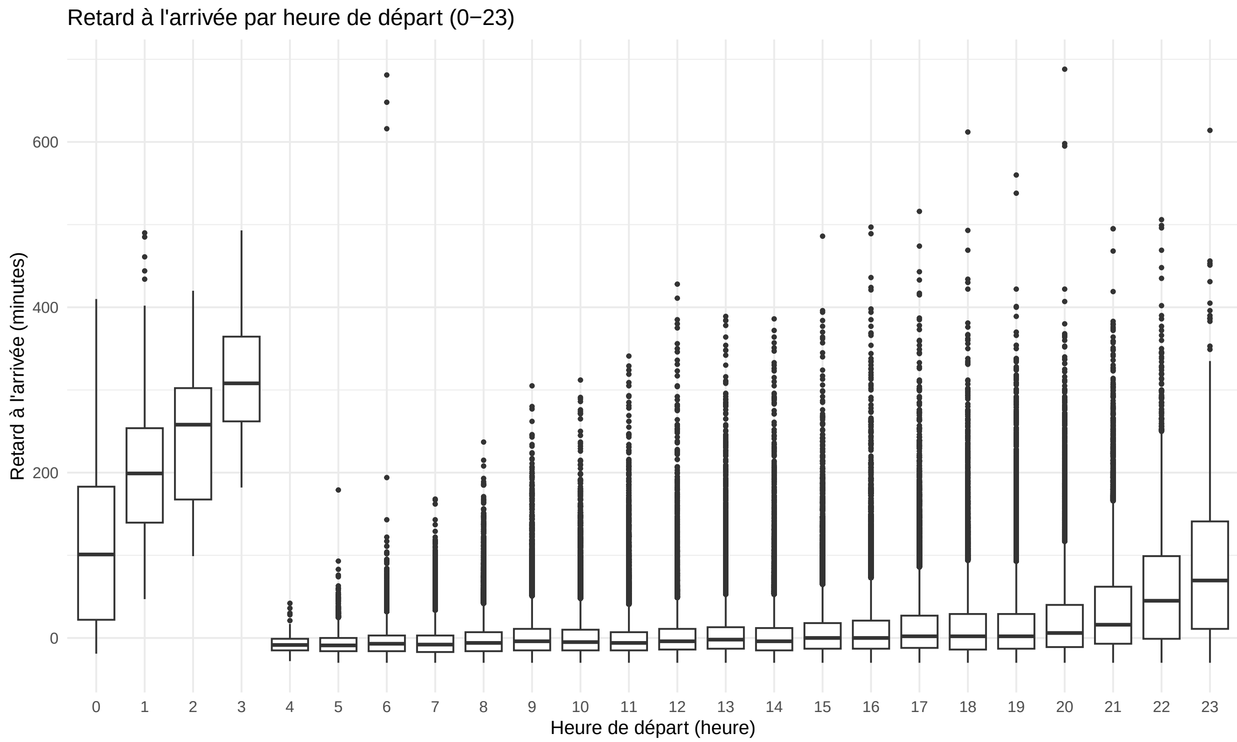
```r
# Affichage du texte avec le nom complet
cat("Aéroport avec le retard moyen le plus faible :", best_dest$name,
    "(", best_dest$dest, ")\n",
    "Retard moyen =", round(best_dest$mean_arr_delay, 2), "min\n",
    "Basé sur", best_dest$n, "vols.\n")
```

```
## Aéroport avec le retard moyen le plus faible : Blue Grass ( LEX )
##  Retard moyen = -22 min
##  Basé sur 1 vols.
```

P. Facteurs de Retard — Heure de Décollage vs Retard

```r
# Créer une variable heure de départ (0-23) et tranche horaire
# dep_time peut être NA; on utilise dep_time (hhmm) converti déjà

df_time <- df_filtered %>%
  mutate(
    dep_hour = floor((as.integer(dep_time) %/% 100) %% 24),
    dep_hour = ifelse(is.na(dep_hour), NA, dep_hour),
    dep_slot = case_when(
      is.na(dep_hour) ~ NA_character_,
      dep_hour >= 5 & dep_hour < 12 ~ "Matin",
      dep_hour >= 12 & dep_hour < 17 ~ "Après-midi",
      dep_hour >= 17 & dep_hour < 22 ~ "Soir",
      TRUE ~ "Nuit"
    )
  )

# Boxplot arr_delay par heure
p5 <- ggplot(df_time %>% filter(!is.na(dep_hour)),
             aes(x = factor(dep_hour), y = arr_delay)) +
```

```r
  geom_boxplot(outlier.size = 0.8) +
  labs(title = "Retard à l'arrivée par heure de départ (0-23)",
       x = "Heure de départ (heure)", y = "Retard à l'arrivée (minutes)") +
  theme_minimal()

print(p5)
```




```r
# Moyennes par tranche horaire
by_slot <- df_time %>% group_by(dep_slot) %>% summarise(n = n(),
                   mean_arr_delay = mean(arr_delay, na.rm = TRUE),
                   sd_arr_delay = sd(arr_delay, na.rm = TRUE), .groups = 'drop')
by_slot %>% kable(digits = 2) %>% kable_styling()
```

dep_slot	n	mean_arr_delay	sd_arr_delay
Après-midi	70023	7.72	37.00
Matin	91266	-1.13	25.06
Nuit	6951	74.08	85.81
Soir	63843	20.53	51.26

```r
# Régression simple : arr_delay ~ dep_hour (modèle cyclique non pris en compte ici)
lm2 <- lm(arr_delay ~ dep_hour, data = df_time %>%
          filter(!is.na(dep_hour) & !is.na(arr_delay)))
summary(lm2)
```

```
##
## Call:
## lm(formula = arr_delay ~ dep_hour, data = df_time %>% filter(!is.na(dep_hour) &
##     !is.na(arr_delay)))
##
```

```
## Residuals:
##    Min     1Q Median     3Q    Max
## -60.66 -22.75  -9.46   8.36 686.54
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -18.32135    0.24586  -74.52   <2e-16 ***
## dep_hour      2.12951    0.01748  121.80   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41.21 on 232081 degrees of freedom
## Multiple R-squared:  0.06008,    Adjusted R-squared:  0.06008
## F-statistic: 1.484e+04 on 1 and 232081 DF,  p-value: < 2.2e-16
```

Q. Compagnies les Plus Retardés

```r
fiabilite_ranked %>%
select(carrier, name, n, mean_arr_delay, prop_delayed_arr, sd_arr_delay) %>%
head(10) %>%
kable(digits = 2) %>% kable_styling()
```

carrier	name	n	mean_arr_delay	prop_delayed_arr	sd_arr_delay
HA	Hawaiian Airlines Inc.	192	-2.60	0.32	26.51
US	US Airways Inc.	14532	3.03	0.39	31.47
AS	Alaska Airlines Inc.	420	3.48	0.39	37.02
DL	Delta Air Lines Inc.	32697	4.20	0.37	39.78
AA	American Airlines Inc.	21620	5.38	0.39	40.06
VX	Virgin America	3204	7.15	0.40	44.56
UA	United Air Lines Inc.	40413	7.19	0.43	38.80
B6	JetBlue Airways	39041	10.71	0.46	41.35
MQ	Envoy Air	18520	11.32	0.48	40.23
WN	Southwest Airlines Co.	8650	11.71	0.47	44.97

R. Analyse de l'Aéroport de Départ

```r
fiabilite_origin <- df_filtered %>%
group_by(origin) %>%
summarise(
n = n(),
mean_arr_delay = mean(arr_delay, na.rm = TRUE),
sd_arr_delay   = sd(arr_delay, na.rm = TRUE),
prop_delayed_dep = mean(dep_delay > 0, na.rm = TRUE),
prop_delayed_arr = mean(arr_delay > 0, na.rm = TRUE),
prop_early_arr   = mean(arr_delay < 0, na.rm = TRUE),
.groups = "drop"
) %>%
arrange(mean_arr_delay)

# Affichage
```

```r
fiabilite_origin %>%
kable(digits = 2) %>%
kable_styling(full_width = FALSE)
```

origin	n	mean_arr_delay	sd_arr_delay	prop_delayed_dep	prop_delayed_arr	prop_early_arr
LGA	72115	8.17	41.72	0.35	0.43	0.55
JFK	75436	8.41	41.19	0.40	0.43	0.56
EWR	84532	12.30	44.17	0.47	0.47	0.51

S. Analyse de Fiabilité par Aéroport d'Arrivée

```r
fiabilite_dest <- df_filtered %>%
group_by(dest) %>%
summarise(
n = n(),
mean_arr_delay = mean(arr_delay, na.rm = TRUE),
sd_arr_delay   = sd(arr_delay, na.rm = TRUE),
prop_delayed_arr = mean(arr_delay > 0, na.rm = TRUE),
prop_early_arr   = mean(arr_delay < 0, na.rm = TRUE),
mean_distance    = mean(distance, na.rm = TRUE),
.groups = "drop"
) %>%
arrange(mean_arr_delay)

# Affichage

fiabilite_dest %>%
kable(digits = 2) %>%
kable_styling(full_width = FALSE)
```

dest	n	mean_arr_delay	sd_arr_delay	prop_delayed_arr	prop_early_arr	mean_distance
LEX	1	-22.00	NA	0.00	1.00	604.00
PSP	16	-9.56	14.16	0.38	0.62	2378.00
STT	387	-0.16	34.67	0.35	0.63	1627.63
SNA	482	0.03	22.74	0.39	0.60	2434.00
HDN	14	2.14	19.79	0.57	0.43	1728.00
SLC	1635	2.86	36.88	0.36	0.62	1986.80
SJU	3974	2.92	33.11	0.37	0.61	1599.85
BOS	10973	3.63	37.66	0.33	0.66	190.70
MIA	8145	4.12	38.61	0.38	0.61	1091.51
PSE	271	4.16	28.78	0.45	0.51	1617.00
MTJ	13	4.38	33.35	0.38	0.62	1795.00
SJC	211	4.79	36.67	0.40	0.58	2569.00
LAS	3879	4.83	36.52	0.41	0.58	2241.17
RSW	2826	4.90	33.44	0.42	0.56	1072.90
HNL	437	5.06	38.67	0.40	0.57	4971.79
SEA	2290	5.22	38.97	0.39	0.59	2412.36
LAX	10435	5.75	36.95	0.43	0.55	2468.77
LGB	429	5.83	36.03	0.42	0.57	2465.00
MVY	65	6.02	39.80	0.32	0.66	173.00

EYW	17	6.35	20.76	0.59	0.35	1207.00
DFW	5664	6.36	41.19	0.40	0.58	1383.05
SRQ	892	6.47	37.91	0.43	0.55	1044.48
DTW	6482	6.84	42.07	0.38	0.60	498.07
MYR	54	6.85	44.29	0.37	0.63	550.24
PHX	3160	7.08	37.39	0.46	0.52	2141.30
MCO	10132	7.17	40.46	0.43	0.56	943.02
OAK	189	7.26	37.65	0.41	0.58	2576.00
AVL	165	7.50	30.95	0.45	0.51	583.87
SFO	8208	7.77	41.19	0.43	0.55	2578.20
BQN	666	7.98	34.17	0.46	0.53	1579.03
IAH	5044	8.04	38.27	0.46	0.52	1407.02
SAN	1749	8.30	35.64	0.48	0.50	2437.27
PDX	797	8.36	39.54	0.47	0.51	2445.77
PIT	2003	8.52	42.66	0.41	0.58	334.05
CLT	9955	8.68	39.67	0.45	0.53	537.90
ACK	92	8.87	34.39	0.40	0.58	199.00
TPA	5459	9.20	41.09	0.44	0.54	1003.81
CLE	3276	9.47	43.38	0.41	0.57	414.00
BGR	231	9.71	45.48	0.41	0.58	378.00
SYR	1248	9.72	39.36	0.41	0.57	205.96
BTV	1913	9.83	39.21	0.42	0.56	265.45
BUR	256	9.85	38.20	0.50	0.48	2465.00
MSP	4879	9.88	44.45	0.44	0.54	1017.45
AUS	1677	10.01	41.47	0.46	0.53	1514.52
PBI	5037	10.33	40.76	0.47	0.51	1028.71
DCA	6744	10.38	39.72	0.45	0.53	210.88
ORD	11216	10.57	46.92	0.42	0.56	728.94
ATL	12471	10.58	43.14	0.48	0.50	757.20
BUF	3376	10.78	42.85	0.42	0.56	296.33
MSY	2569	10.82	44.45	0.46	0.53	1177.67
ABQ	144	10.85	36.87	0.50	0.49	1826.00
FLL	8851	11.04	42.10	0.48	0.51	1069.96
CMH	2425	11.04	41.23	0.45	0.53	476.75
BDL	324	11.35	43.80	0.41	0.58	116.00
HOU	1415	11.38	41.97	0.50	0.49	1420.03
DEN	4975	11.49	42.33	0.49	0.49	1614.70
XNA	713	11.71	39.27	0.52	0.47	1142.33
ROC	1752	11.76	47.41	0.41	0.58	259.36
RDU	5801	11.93	43.45	0.45	0.53	426.57
EGE	183	12.13	46.45	0.49	0.49	1736.46
MDW	2961	12.98	45.55	0.48	0.50	718.00
IND	1485	13.48	47.08	0.46	0.52	651.98
BWI	1268	13.75	49.13	0.43	0.56	178.33
PHL	1100	13.77	45.58	0.47	0.51	94.11
STL	3027	13.78	46.26	0.48	0.50	878.79
SMF	179	13.92	40.31	0.56	0.44	2521.00
TVC	14	14.00	41.11	0.36	0.57	655.00
PWM	1697	14.03	44.59	0.46	0.53	276.54
CHS	1884	14.18	45.53	0.48	0.50	632.64

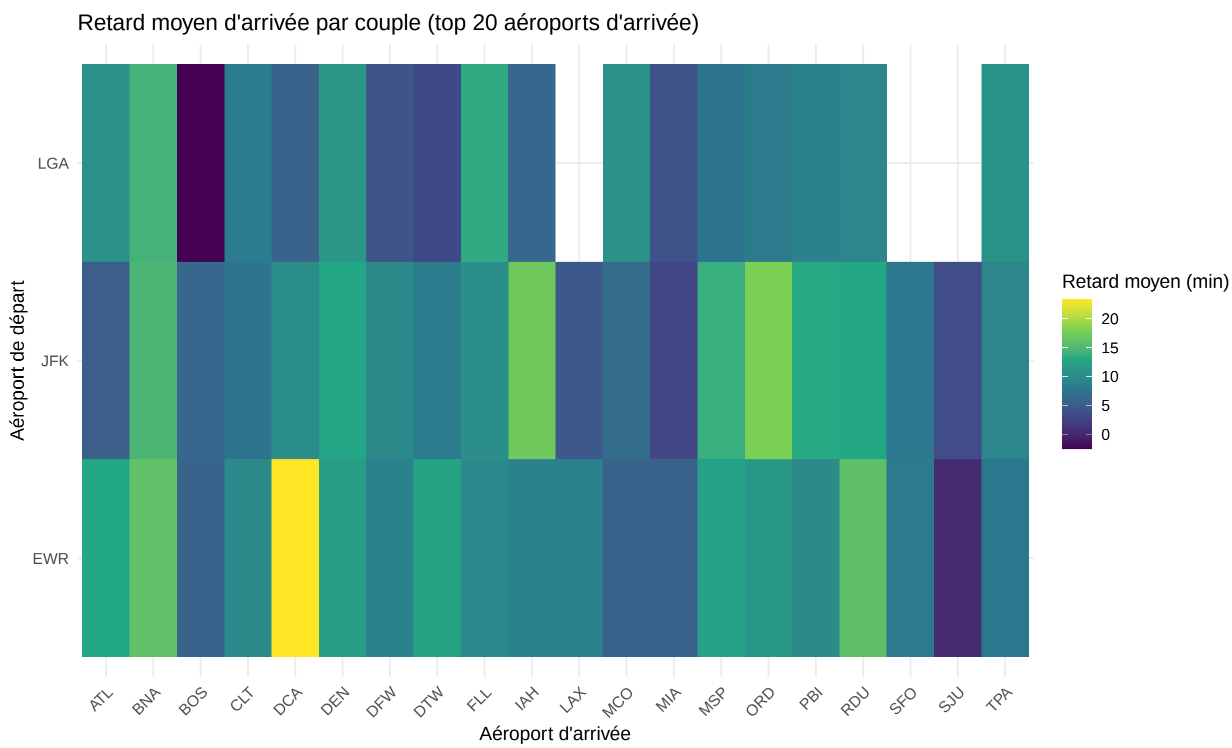
MEM	1183	14.29	43.62	0.50	0.49	954.03
ILM	75	14.43	44.83	0.47	0.52	500.00
CHO	24	14.46	57.62	0.38	0.62	305.00
JAX	1878	14.48	43.17	0.51	0.47	824.72
SAT	422	14.92	58.36	0.47	0.52	1577.74
MHT	697	14.98	43.08	0.45	0.53	208.08
BNA	4356	14.98	48.03	0.48	0.50	758.22
IAD	3950	15.25	50.39	0.46	0.52	224.68
ORF	1067	15.29	49.93	0.47	0.51	288.49
CRW	131	15.76	42.55	0.49	0.50	444.00
MKE	1978	16.45	48.31	0.52	0.46	733.10
BZN	19	17.05	41.00	0.58	0.42	1882.00
SDF	782	17.12	44.52	0.52	0.47	645.18
ALB	365	17.67	52.09	0.48	0.51	143.00
CVG	2690	17.79	51.96	0.48	0.50	575.02
CAK	605	18.17	54.38	0.54	0.44	397.00
GSO	1069	18.22	48.27	0.50	0.48	449.03
PVD	312	18.55	47.17	0.52	0.47	160.00
SAV	546	18.64	52.80	0.51	0.47	708.90
DAY	1008	19.01	49.24	0.52	0.47	536.24
BHM	179	21.17	55.01	0.50	0.48	865.99
GRR	558	21.22	49.90	0.57	0.40	605.89
GSP	611	21.53	48.83	0.54	0.44	595.75
OMA	547	21.67	51.96	0.56	0.42	1135.46
MCI	1309	21.75	52.61	0.56	0.42	1096.97
RIC	1683	24.80	54.64	0.55	0.43	280.83
MSN	417	25.05	58.14	0.56	0.43	803.93
DSM	341	27.01	57.29	0.58	0.41	1019.71
TYS	426	27.45	56.02	0.57	0.41	638.14
JAC	21	28.10	41.03	0.81	0.14	1875.90
SBN	4	30.50	25.49	0.75	0.00	647.50
OKC	231	37.16	52.47	0.72	0.27	1325.00
TUL	213	37.84	57.90	0.73	0.26	1215.00
CAE	84	38.81	49.71	0.81	0.17	604.14

U. Couple Aéroport de Départ/Aéroport d'Arrivée

```
couples_retards <- df_filtered %>%
group_by(origin, dest) %>%
summarise(
n = n(),
mean_arr_delay = mean(arr_delay, na.rm = TRUE),
sd_arr_delay = sd(arr_delay, na.rm = TRUE),
prop_delayed = mean(arr_delay > 0, na.rm = TRUE),
.groups = "drop"
) %>%

filter(n >= 30) %>%
arrange(desc(mean_arr_delay))
```

```
couples_retards %>%
head(15) %>%
kable(digits = 2) %>%
kable_styling(full_width = FALSE)
```

origin	dest	n	mean_arr_delay	sd_arr_delay	prop_delayed
EWR	TYS	236	42.57	64.22	0.67
EWR	CAE	72	42.00	51.18	0.83
EWR	TUL	213	37.84	57.90	0.73
EWR	OKC	231	37.16	52.47	0.72
EWR	DSM	275	29.69	60.10	0.58
LGA	OMA	57	29.21	56.71	0.54
EWR	RIC	1205	28.64	54.28	0.60
EWR	MSN	259	28.57	59.00	0.61
EWR	PWM	620	25.94	52.45	0.59
EWR	MKE	799	25.68	52.83	0.61
EWR	MCI	929	25.59	52.59	0.61
EWR	DCA	1257	23.28	46.41	0.61
EWR	GSP	544	23.06	50.24	0.55
JFK	CVG	639	23.03	58.04	0.51
EWR	DAY	804	22.99	49.88	0.56



V. Nombre de Vols avec au moins 2h de Retard

```
strict_cases <- df_master %>%
filter(!is.na(arr_delay), !is.na(dep_delay)) %>%
```

```r
filter(arr_delay > 120,
dep_delay <= 0)

cat("Vols arrivés avec >2h de retard mais partis à l'heure:", nrow(strict_cases), "\n")
```

```
## Vols arrivés avec >2h de retard mais partis à l'heure: 164
```

3. Analyse des Annulations et Données Manquantes

A. Audits des Données Manquantes

Calcul de la proportion de NA pour les variables critiques

```r
missing_stats <- flights %>%
  summarise(
    na_dep_time = mean(is.na(dep_time)),
    na_dep_delay = mean(is.na(dep_delay)),
    na_arr_time = mean(is.na(arr_time)),
    na_arr_delay = mean(is.na(arr_delay))
  ) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "taux_na")

missing_stats
```

```
## # A tibble: 4 x 2
##   variable     taux_na
##   <chr>          <dbl>
## 1 na_dep_time   0.0256
## 2 na_dep_delay  0.0256
## 3 na_arr_time   0.0268
## 4 na_arr_delay  0.0288
```

Identification des autres colonnes concernées par les manquants

```r
col_na_counts <- colSums(is.na(flights))
col_na_counts[col_na_counts > 0]
```

```
##  dep_time dep_delay  arr_time arr_delay   tailnum  air_time
##      6481      6481      6767      7273      1973      7273
```

B. Analyse des Vols Annulés

Définition : Un vol est annulé si dep_time ET arr_time sont manquants

```r
cancelled_flights <- flights %>%
  filter(is.na(dep_time) & is.na(arr_time))

# Nombre total
nb_cancelled <- nrow(cancelled_flights)
print(paste("Nombre total de vols annulés :", nb_cancelled))
```

```
## [1] "Nombre total de vols annulés : 6481"
```

Affinage par Destination & Compagnie

```r
# Top 5 des destinations subissant le plus d'annulations
cancelled_flights %>%
  count(dest, sort = TRUE) %>%
  head(5) %>%
  left_join(airports, by = c("dest" = "faa")) %>%
  select(Destination = dest, Ville = name, Annulations = n) %>%
  knitr::kable(caption = "Top 5 Destinations : Les plus touchées par les annulations") %>%
  kableExtra::kable_styling(full_width = FALSE)
```

Table 11: Top 5 Destinations : Les plus touchées par les annulations

Destination	Ville	Annulations
ORD	Chicago Ohare Intl	475
DCA	Ronald Reagan Washington Natl	435
BOS	General Edward Lawrence Logan Intl	351
RDU	Raleigh Durham Intl	284
CLT	Charlotte Douglas Intl	264

```r
# Classement par Compagnie
cancelled_by_carrier <- cancelled_flights %>%
  count(carrier, sort = TRUE) %>%
  left_join(airlines, by = "carrier") %>%
  select(Code = carrier, Compagnie = name, Annulations = n)

cancelled_by_carrier
```

```
## # A tibble: 14 x 3
##    Code  Compagnie                   Annulations
##    <chr> <chr>                             <int>
##  1 EV    ExpressJet Airlines Inc.           2183
##  2 MQ    Envoy Air                           936
##  3 9E    Endeavor Air Inc.                   786
##  4 UA    United Air Lines Inc.               563
##  5 US    US Airways Inc.                     513
##  6 AA    American Airlines Inc.              503
##  7 B6    JetBlue Airways                     405
##  8 DL    Delta Air Lines Inc.                285
##  9 WN    Southwest Airlines Co.              179
## 10 FL    AirTran Airways Corporation          56
## 11 YV    Mesa Airlines Inc.                   40
## 12 VX    Virgin America                       28
## 13 AS    Alaska Airlines Inc.                  2
## 14 F9    Frontier Airlines Inc.                2
```

Visualisation des Annulations par Compagnie

```r
cancelled_flights %>%
  count(carrier, sort = TRUE) %>%
  left_join(airlines, by = "carrier") %>%
  ggplot(aes(x = reorder(name, n), y = n)) +
  geom_col(fill = "firebrick") +
```

```r
  coord_flip() +
  labs(
    title = "Nombre de vols annulés par Compagnie Aérienne",
    subtitle = "Critère : Heures de départ et d'arrivée manquantes",
    x = "Compagnie",
    y = "Nombre d'annulations"
  ) +
  theme_minimal()
```